Theory Note 4 of 8

Nmap Scan Types and What They Leave Behind

Network Traffic Analysis, Nmap and Windows Logging | Week 2

1. Overview

Nmap is the standard tool for finding live hosts and open ports. Each scan option works differently, and each leaves its own footprint in network traffic and in logs. Knowing those footprints is useful on both sides: a tester can pick the right scan, and a defender can recognise it in Wireshark, firewall logs or an IDS. These scans should only be run on systems you own or have written permission to test.

2. Scan types compared

Option	How it works	Footprint in packets	Footprint in logs
-sn Ping scan	Finds live hosts without scanning ports. On the local network it uses ARP requests. Across networks it sends an ICMP echo, a TCP SYN to port 443, a TCP ACK to port 80 and an ICMP timestamp request.	Many ARP who-has requests across the subnet, or ICMP echo requests to a range of addresses.	Firewall and IDS ping-sweep alerts. Little on the hosts unless ICMP logging is on.
-sS SYN (half-open) scan	Sends a SYN. A SYN-ACK means open, and Nmap answers with a RST rather than finishing the handshake. A RST means closed. No reply means filtered. This is the default scan for privileged users.	A burst of SYN packets from one source to many ports, followed by RSTs, and no final ACK. Filter: tcp.flags.syn==1 && tcp.flags.ack==0	Often missed by application logs because no full connection forms. Visible in firewall logs and in IDS tools such as Snort.
-sT TCP connect scan	Uses the operating system's connect() call, so the full handshake completes and is then closed. Used when raw sockets are not available.	Complete SYN, SYN-ACK, ACK sequences followed by FIN or RST for every open port.	Services record the connections (web server, SSH, and so on), so it is easier to spot in application logs.
-sU UDP scan	Sends empty or protocol-specific UDP datagrams. An ICMP port-unreachable reply means closed, a UDP reply means open, and silence means open or filtered. It is slow because of ICMP rate limiting.	Many UDP packets to different ports and many ICMP type 3, code 3 replies.	Service logs for DNS, SNMP or DHCP may show empty or malformed requests, plus firewall drops.
-sV Version detection	After finding open ports, it connects and sends probes, then compares the replies with the nmap-service-probes database to identify the service and version.	Full connections carrying unusual probe payloads, such as odd HTTP requests or varied TLS hellos.	Junk requests, 400 errors and banner grabs in application and web server logs.
-O OS detection	Sends crafted TCP, UDP and ICMP probes and compares details of the replies (TTL, window size, TCP options) with a fingerprint database. Works best with one open and one closed port.	Unusual TCP option combinations and probes with odd flags.	Mostly IDS alerts for OS fingerprinting. Very little on the host itself.
-A Aggressive scan	A shortcut that combines -sV, -O, default NSE scripts (-sC) and traceroute.	All of the above together, plus script traffic and TTL-limited traceroute packets.	The noisiest option, producing many alerts and log entries across web, SMB and SSH services.

3. Detection notes

- A port scan looks like one source touching many ports or many hosts in a short time.
- The SYN scan is quieter for the target host, but on the network it is plain to see as SYN followed by RST.
- Results can be saved with -oN (normal text) and -oX (XML). The -oA option writes every format at once.
- For the Week 3 IDS work, the SYN scan pattern is the one to build a detection around.

> **Key takeaway**
> The more detail a scan collects, the more noise it creates. Ping scans are light, SYN scans are quiet on the host but visible on the wire, and -A is loud everywhere.